

Mohamed Elsayed

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Education

Doctoral of Philosophy

University of Alberta, Canada
Computing Science

Expected Graduation: Sep 2026

Master of Science

University of Alberta, Canada
Computing Science [Thesis-based]

GPA: 3.9/4.0

2019-2022

Bachelor of Science

University of science and technology, Zewail City, Egypt

GPA: 3.7/4.0

2014 – 2019

Major: Communication and Information Engineering. Minor: Physics of the Earth and Universe.

Experience

Huawei Technologies Co., Ltd

Support Researcher, Intern

Edmonton, Canada

May 2020 - Dec 2020

- Worked on autonomous intersection navigation for the unprotected left turn problem with RL.
- Co-authored a workshop paper in NeuroIPS ML4AD.

Nara Institute of Science and Technology

Robotics Research Intern

Nara, Japan

Aug 2018 - Sep 2018

- Worked on Semantic SLAM problem using deep learning techniques.
- Worked on map in-painting techniques.

Valeo

Deep Learning Research Intern

Smart Village, Egypt

Jul 2018 - Aug 2018

- Worked on automotive-related projects (classification, semantic segmentation, and detection).
- Handled new state-of-the-art network architectures and bench-marked on related works.

ONE Lab

Machine Learning Research Intern

Cairo, Egypt

Jan 2018 - Mar 2018

- Developed features that classify EEG signals into ictal and non-ictal signals for epilepsy patients.
- Published the results and findings in a conference paper bench-marking with related works.

Teaching

University of Alberta

Graduate Teaching Assistant

Edmonton, Canada

Sep 2019 - April 2020, Sep 2022 - Dec 2022

- Taught two classes: introduction to programming and introduction to reinforcement learning.

Zewail City of Science and Technology

Junior Teaching Assistant

Cairo, Egypt

Sep 2018 - Dec 2018

- Conducted presentations and assessed students in their assignments and problems in ML class.

Selected Projects

RL Algorithms on Real-world Robots

Oct 2019 – Sep 2020

- Improved docking and moving tasks developed by Kindred.AI.
- Produced robust docking in Roomba with online RL.

Autonomous Robot Navigation in Dynamic Environment

Dec 2018 – May 2019

- Replicated and improved results from a journal paper that used deep learning in navigation.
- Developed hardware implementation using Jetson Nano, camera, and RPLiDAR.
- Developed full simulation for the system using Gazebo and Rviz.

Robot Navigation - Simulation and Implementation

Jan 2018 - Mar 2018

- Develop a full simulation for simultaneous localization and mapping based on Gazebo and ROS.
- Transformed the simulation with real hardware using Raspberry Pi 3 and Tiva C.

Extracurricular Activities

- **Reviewer** at TMLR, ICLR 2023 and NeurIPS 2022.
- **Team Leader** in MBZIRC team (2018) [submitted proposal].
- **President** of IEEE Zewail City Student Branch. (2017)
Being in this position for a year has given me good experience in building and working with teams as a leader in addition to some managerial and planning skills (Granted University Best Leader Award).
- **Director** of Walking Robots Competition in Zewail City. (2017)
- Vice-president of IEEE Zewail City Student Branch. (2016)
- **Content Developer** at Zewail City Science Festival. (2016)
- **Head** of Organizing Committee of ICMIS14 Conference. (2015)
- **Developer** at Microsoft Club in Zewail City. (2014)

Technical and Personal skills

- **Programming Languages:** Python, C, C++, MATLAB, TeX. Also basic ability with Assembly.
- **Software Skills:** OpenAI Gym, Pytorch, OpenCV, Sci-Kit, ARGoS, ROS, RViz, and Gazebo.
- **Interpersonal Skills:** Leadership, Good presentation skills, Ability to work under pressure, Self-learning, Works well in a team.
- **Platforms:** Linux, Windows.
- **Relevant Coursework:** Reinforcement Learning I and II, Reinforcement Learning with Robots, Autonomous Robot Navigation, Data Structures and Algorithm Analysis, Computer Vision, Machine Learning, Operating Systems, Modern Control, Embedded systems, Computer Architecture, Linear Algebra, Discrete Math, Probability and Statistics.
- **Languages:** Arabic (Mother Tongue), English (Very good).

Publications

Conferences and Journals:

1. Mohamed A. Elgammal, Omar A. Elkhoully, Heba Elhosary, **Mohamed Elsayed**, Ahmed Nader Mohieldin, Khaled N. Salamak, Hassan Mostafa, Linear and Nonlinear Feature Extraction for Neural Seizure Detection, *IEEE International Midwest Symposium on Circuits and Systems*, 2018.
2. Ming Zhou, Jun Luo, Julian Vilella, Yaodong Yang, David Rusu, Jiayu Miao, Weinan Zhang, Montgomery Alban, Iman Fadarar, Zheng Chen, Aurora Chongxi Huang, Ying Wen, Kimia Hassanzadeh, Daniel Graves, Dong Chen, Zhengbang Zhu, Nhat Nguyen, **Mohamed Elsayed**, Kun Shao, Sanjeevan Ahilan, Baokuan Zhang, Jiannan Wu, Zhengang Fu, Kasra Rezaee, Peyman Yadmellat, Mohsen Rohani, Nicolas Perez Nieves, Yihan Ni, Seyedershad Banijamali, Alexander Cowen Rivers, Zheng Tian, Daniel Palenicek, Haitham bou Ammar, Hongbo Zhang, Wulong Liu, Jianye Hao, Jun Wang. SMARTS: Scalable Multi-Agent Reinforcement Learning Training School for Autonomous Driving. *Conference on Robot Learning (CoRL)*, 2020.

Workshops:

1. **Mohamed Elsayed**, A. Rupam Mahmood. HesScale: Scalable Computation of Hessian Diagonals. *NeurIPS Workshop on Higher-Order Optimization in Machine Learning*, 2022.
2. **Mohamed Elsayed***, Kimia Hassanzadeh*, Nhat Nguyen*, Montgomery Alban, Xiru Zhu, Daniel Graves, Jun Luo, ULTRA: A Reinforcement Learning Generalization Benchmark for Autonomous Driving, *NeurIPS Workshop on Machine Learning for Autonomous Driving (ML4AD)*, 2020.
3. **Mohamed Elsayed**, Hager Radi, A. Rupam Mahmood, Learning to Dock Robustly, *ICML Workshop in Reinforcement Learning for Real Life (RL4RealLife)*, 2020.

Pre-prints:

1. **Mohamed Elsayed**, A. Rupam Mahmood (2022). HesScale: Scalable Computation of Hessian Diagonals. *arXiv preprint arXiv:2210.11639*.
2. **Mohamed Elsayed**, A. Rupam Mahmood (2023). Utility-based Perturbed Gradient Descent: An Optimizer for Continual Learning. *arXiv preprint arXiv:2302.03281*.

References

Available Upon Request